Week 09 — Multivariate Time Series & VAR

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The big picture. Move to vectors: matrix autocovariance (the lag is in the FIRST argument!), spectral matrix, coherence, and the VAR model with its companion form.

We now observe several series at once: a basket of stocks (Apple, Nasdaq-100, S&P 500), or the position and temperature of an ocean drifter. Each X_t becomes a *vector* in $\mathbb{R}^d$. Two questions: how do we *describe* the second-order structure (now matrix-valued, with genuine **cross**-terms between components), and how do we *model* it (answer: the VAR). The recurring trick: *every VAR(p) is a VAR(1) in disguise*, so the whole theory reduces to a single matrix.

Key definitions

Definition — Multivariate series & second-order stationarity

$\{X_t\}$ with $X_t = (X_t^{(1)}, \dots, X_t^{(d)})^\top \in \mathbb{R}^d$, each marginal a univariate series. It is **(weakly) second-order stationary** if the mean is constant, $\mathrm{tr}(\mathrm{Var}(X_t)) < \infty$, and

$$\mathrm{Cov}(X_{t+\tau}, X_t) \text{ depends only on } \tau.$$

Equivalently: each marginal is stationary *and* every pair has a shift-invariant cross-covariance.

Definition — Matrix autocovariance sequence (ACVS) & cross-correlation

$$\Gamma_\tau = \mathrm{Cov}(X_{t+\tau}, X_t), \qquad \gamma_\tau^{(j,k)} = \mathrm{Cov}\big(X_{t+\tau}^{(j)}, X_t^{(k)}\big).$$

The lag τ sits in the FIRST argument (irrelevant in the scalar case, crucial here). Diagonal ($j = k$): the usual marginal ACVS. Off-diagonal ($j \neq k$): the **cross-covariance**, which encodes lead–lag relations between channels. The **cross-correlation sequence** (CCS) is

$$\rho_\tau^{(j,k)} = \frac{\gamma_\tau^{(j,k)}}{\sqrt{\gamma_0^{(j,j)}\,\gamma_0^{(k,k)}}}, \qquad \rho_\tau^{(j,k)} = \rho_{-\tau}^{(k,j)}.$$

Definition — Multivariate white noise $\mathrm{WN}(0,\Sigma)$

$\{\varepsilon_t\} \sim \mathrm{WN}(0,\Sigma)$ iff it is stationary, mean zero, with $\Gamma_\tau^{(\varepsilon)} = \Sigma$ at $\tau = 0$ and 0 otherwise. Components are uncorrelated **across time** but may be correlated **contemporaneously** (off-diagonal entries of Σ). One sometimes assumes Σ diagonal — an *extra* assumption, not part of the definition.

Definition — Spectral density matrix

If all cross-covariances are summable (ℓ^1),

$$S(f) = \sum_{\tau\in\mathbb{Z}} \Gamma_\tau\, e^{-2\pi i\tau f}, \qquad \Gamma_\tau = \int_{-1/2}^{1/2} S(f)\, e^{2\pi i f\tau}\, df, \qquad f \in [-\tfrac{1}{2}, \tfrac{1}{2}].$$

Diagonal: the usual (real) spectral density $\mathcal{S}_j(f)$. Off-diagonal: the **cross-spectrum** $S_{j,k}(f) \in \mathbb{C}$ (complex in general).

Definition — Coherence

$$r_{j,k}(f) = \frac{S_{j,k}(f)}{\sqrt{S_{j,j}(f)\, S_{k,k}(f)}} \in \mathbb{C}.$$

The frequency-by-frequency "correlation" between channels (literally the correlation of the increments dZ_j, dZ_k below). Report its **magnitude** $|r_{j,k}(f)|$ (strength of association, $|r|^2 \in [0,1]$) and its **phase** $\arg r_{j,k}(f)$ (the "sign" / lead–lag angle).

Definition — VAR(p) & companion form

With $\{\varepsilon_t\} \sim \mathrm{WN}(0,\Sigma)$ and $\Phi_p \neq 0$,

$$X_t = \Phi_1 X_{t-1} + \cdots + \Phi_p X_{t-p} + \varepsilon_t, \qquad \Phi(B)X_t = \varepsilon_t, \quad \Phi(z) = I_d - \sum_{j=1}^{p} \Phi_j z^j.$$

Stacking $Y_t = (X_t^\mathsf{T}, \ldots, X_{t-p+1}^\mathsf{T})^\mathsf{T} \in \mathbb{R}^{dp}$ turns it into a VAR(1), $Y_t = F\, Y_{t-1} + U_t$, with

$$F = \begin{pmatrix} \Phi_1 & \Phi_2 & \cdots & \Phi_{p-1} & \Phi_p \\ I_d & 0 & \cdots & 0 & 0 \\ 0 & I_d & & & \vdots \\ \vdots & & \ddots & & \vdots \\ 0 & \cdots & & I_d & 0 \end{pmatrix}, \quad U_t = \begin{pmatrix} \varepsilon_t \\ 0 \\ \vdots \\ 0 \end{pmatrix}.$$

Recover the original process via $X_t = G\, Y_t$ with $G = (I_d \; 0 \; \cdots \; 0)$. The off-diagonal entries of the Φ_j *are* the cross-component dynamics.

Key results & formulas

Symmetry of the ACVS and the spectrum

$$\gamma_\tau^{(j,k)} = \gamma_{-\tau}^{(k,j)} \iff \Gamma_{-\tau} = \Gamma_\tau^\mathsf{T}, \qquad S(f) = S(f)^\mathsf{H} = \overline{S(-f)^\mathsf{T}}, \quad S(f) \succeq 0.$$

$\{\Gamma_\tau\}$ is positive semi-definite: $\sum_{j,k} a_j^\mathsf{T} \Gamma_{k-j} a_k \geq 0$ (it is a variance). **Warning:** off-diagonal, the ACVS is *not* even in τ; you must swap the indices $(j,k) \to (k,j)$ when you flip the sign of τ. This asymmetry is exactly what encodes lead–lag.

Theorem — Multivariate spectral representation

There is a vector orthogonal-increment process $\{Z(f)\}$ with

$$X_t = \mu + \int_{-1/2}^{1/2} e^{2\pi i f t}\, dZ(f),$$
$$\mathbb{E}[dZ(f)] = 0, \quad \mathrm{Var}\left(dZ(f)\right) = S(f)\, df, \quad \mathrm{Cov}\left(dZ(f), dZ(f')\right) = 0 \;(f \neq f').$$

Idea: the vector analogue of Cramér. So a stationary vector process is a sum of uncorrelated random sinusoids whose (matrix) variance is $S(f)\, df$; coherence is literally the correlation of $dZ_j(f), dZ_k(f)$.

Stability of a VAR(p)

Stable iff the roots of the reverse characteristic polynomial $\det(I_d - \Phi_1 z - \cdots - \Phi_p z^p) = 0$ *all* lie outside the unit circle ($|z| > 1$); equivalently iff F has all eigenvalues $|\lambda| < 1$. Stability $\Rightarrow$ stationarity (not conversely in general). **VAR(1) shortcut:** just check $|\lambda_i(\Phi_1)| < 1$ (spectral radius < 1); for triangular Φ_1 these are the diagonal entries. Sanity check: $\det(\cdot)\big|_{z=0} = \det I_d = 1$.

MA(∞) and covariance of a stable VAR(1)

$$X_t = \sum_{j\geq 0} \Phi_1^j \, \varepsilon_{t-j}, \qquad \Gamma_\tau = \sum_{k\geq 0} \Phi_1^{k+\tau} \, \Sigma \, (\Phi_1^k)^\top \quad (\tau \geq 0).$$

At $\tau = 0$ this is the discrete Lyapunov equation $\Gamma_0 = \Phi_1 \Gamma_0 \Phi_1^\top + \Sigma$. For a VAR($p$): same formula with F, Σ_U (top-left block Σ), then project $\Gamma_\tau^{(X)} = G\big(\sum_k F^{k+\tau} \Sigma_U (F^k)^\top\big) G^\top$.

Theorem — Yule–Walker equations (VAR)

For $\tau \geq 0$,

$$\Gamma_\tau = \Phi_1 \Gamma_{\tau-1} + \cdots + \Phi_p \Gamma_{\tau-p} + \delta_{\tau,0}\, \Sigma.$$

Idea: multiply the VAR equation on the right by $X_t^\top$ and take expectations; the noise term contributes Σ only at $\tau = 0$. Solve the first $p+1$ equations for $\Gamma_0, \ldots, \Gamma_p$, then recurse for all $\tau > p$. Run in reverse, it *estimates* the Φ_j from sample covariances (VAR fitting).

Two signatures: common noise & pure delay

Common noise. If $Y_t = X_t + \phi_t \mathbf{1}$ (the *same* scalar noise in both channels), then $\Gamma_\tau^{(Y)} = \Gamma_\tau^{(X)} + \sigma_\phi^2 \, \delta_{\tau,0} \, \mathbf{1}\mathbf{1}^\top$: correlation added *only* at lag 0.
Pure delay. If $X_t = (Y_t, Y_{t-\nu})^\top$, then $S_{1,2}(f) = \mathcal{S}_Y(f)\, e^{2\pi i f \nu}$, hence coherence of magnitude 1 and phase $2\pi\nu f$ (*linear* in f).

Intuition

Why the lag lives in the first argument

Scalar: $\gamma_\tau = \gamma_{-\tau}$, so argument order is irrelevant. Multivariate: it becomes *crucial* — it fixes the convention $\Gamma_{-\tau} = \Gamma_\tau^\top$. Diagonals stay even, but a cross-covariance $\gamma_\tau^{(1,2)}$ (channel 1 leading channel 2 by τ) has no reason to equal $\gamma_{-\tau}^{(1,2)}$. Classic error: forgetting to swap (j,k) when you flip τ.

The noise only shows up at $\tau = 0$

X_t is built from $\varepsilon_t, \varepsilon_{t-1}, \ldots$ (the causal MA(∞)). For $\tau > 0$, $\varepsilon_{t+\tau}$ lies in the *future* of X_t, hence uncorrelated: $\mathrm{Cov}(\varepsilon_{t+\tau}, X_t) = 0$. Only $\tau = 0$ gives $\mathrm{Cov}(\varepsilon_t, X_t) = \Sigma$. Exactly the scalar Yule–Walker mechanism, transposed to matrices.

"Everything is a VAR(1)"

The companion form collapses any VAR(p) into a single VAR(1) on F. So you prove the VAR(1) theory *once* (stability, MA(∞), covariances), then reuse it for every p via $X_t = GY_t$. The top block row of F carries the real equation; the lower I_d blocks just shift the stack.

Delay $\leftrightarrow$ linear phase

A shift of ν samples in time = a factor $e^{\pm 2\pi i \nu f}$ in the spectrum = a *linear* phase $\mp 2\pi\nu f$. A coherence of magnitude 1 whose phase grows proportionally to f (slope $2\pi\nu$) is the textbook signature of a lead/lag of ν samples between two channels.

Where the channel couplings live

The *off-diagonal* entries of the Φ_j *and* of Σ are the only links between components. If Φ_1 and Σ are diagonal, the VAR(1) decouples into d independent scalar AR(1)'s ($\Gamma_0^{(j,j)} = \sigma_j^2/(1-\phi_j^2)$): cross-spectrum zero, coherence zero everywhere.

Key takeaways

Key takeaways — the week's reflexes

- **Lag in the first argument** $\Rightarrow \Gamma_{-\tau} = \Gamma_\tau^\mathsf{T}$ and $\gamma_\tau^{(j,k)} = \gamma_{-\tau}^{(k,j)}$: always swap indices when flipping τ.
- **Test stability:** roots of $\det(I_d - \sum \Phi_j z^j)$ all $|z| > 1$, or eigenvalues of F all $|\lambda| < 1$. Sanity check: $\det(\cdot)\big|_{z=0} = 1$. For VAR(1), read off $\lambda_i(\Phi_1)$ directly.
- **MA(∞) & covariance:** write $X_t = \sum \Phi_1^j \varepsilon_{t-j}$, get $\Gamma_\tau = \sum_k \Phi_1^{k+\tau}\Sigma(\Phi_1^k)^\mathsf{T}$, or solve Lyapunov for Γ_0.
- **Yule–Walker recursion:** $\Gamma_\tau = \sum_i \Phi_i \Gamma_{\tau-i}$ (add Σ at $\tau = 0$); mind the transpose $\Gamma_{-s} = \Gamma_s^\mathsf{T}$.
- **Cross-spectrum:** find $\gamma_\tau^{(j,k)}$ (often a *shift* of a known covariance), Fourier-transform, re-index to pull out $e^{\pm 2\pi i \nu f}$, read magnitude & phase, divide by $\sqrt{S_{jj}S_{kk}}$ for coherence.
- **Signature reflexes:** common noise $\to$ correlation at lag 0 only; pure delay $\to$ coherence 1, linear phase.